

RetailIQ

Instacart Market Basket Analysis

End-to-End Retail Analytics Project

3M Total Orders	206K Total Customers	50K Total Products	58.97% Reorder Rate
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Prepared by	Anirban Sur
Tools Used	Python SQL Power BI
Dataset	Instacart Market Basket Analysis

Executive Summary & Business Objectives

Project Summary

RetailIQ is an end-to-end retail analytics project built on the Instacart Market Basket Analysis dataset. The project follows a complete analytics workflow — from business and data understanding through data cleaning, exploratory analysis, SQL-based business querying, and interactive Power BI reporting — to convert raw grocery-ordering data into decision-ready insight.

Why This Analysis Matters

Online grocery retailers depend on understanding how, when, and what customers order to drive retention, plan inventory, and personalize marketing. By analyzing customer ordering behavior, product reorder patterns, and time-based shopping trends at scale, this project surfaces the kind of operational and customer insight that supports data-driven retail decision-making.

Business Objectives

- Understand customer purchasing behavior and ordering frequency
- Evaluate product performance, popularity, and reorder rates
- Assess department and aisle-level sales contribution
- Analyze time-based ordering patterns, including weekday vs. weekend behavior
- Translate raw transactional data into a governed, reportable data model
- Deliver actionable business recommendations through an interactive dashboard

How Python, SQL & Power BI Were Used

Tool	Role in the Project
Python (Pandas)	Data quality assessment, cleaning, and exploratory data analysis (EDA) on the raw Instacart tables
SQL	25+ business-focused queries covering customer, product, department/aisle, time-based, and advanced
Power BI	Data modeling, DAX measures, and a 4-page interactive executive dashboard

Business Value

The project demonstrates how a retail business can move from raw order-level transaction data to a self-service Power BI dashboard that management can use to monitor customer loyalty, product performance, and shopping-time patterns without needing to run manual queries — shortening the path from data to decision.

Data, Data Cleaning & Exploratory Analysis

Dataset Overview

The analysis uses five core tables from the Instacart Market Basket Analysis dataset: **orders**, **order_products__prior**, **products**, **aisles**, and **departments**. The **order_products__train** and **order_products__test** tables, along with the **eval_set** column in **orders**, were excluded to keep the analysis focused on a single, consistent order history spanning 206K customers, 3M orders, and 50K products across 21 departments and 134 aisles.

Data Model

The five tables were structured into a relational model connecting customers to their orders, orders to the products purchased within them, and products to their parent aisle and department — enabling analysis at the customer, order, product, aisle, and department level.

Data Cleaning & Quality Assessment

- Inspected dataset structure and data types across all five tables
- Performed missing value analysis and duplicate checks
- Validated primary and foreign key relationships between tables
- Checked cross-table data consistency prior to analysis

Exploratory Data Analysis

EDA was performed in Python covering customer-level behavior, product-level performance, department and aisle contribution, and time-based ordering patterns. These findings informed the metrics and visuals later built into the Power BI dashboard, including basket size, reorder rate, and peak ordering times discussed in the following sections.

SQL Business Analysis

A set of 25+ business-focused SQL queries was written against the cleaned data model to answer core retail questions — from basic operational metrics to multi-step behavioral analysis. The queries are organized into five categories:

Category	Business Questions Answered
Basic Business Metrics	Overall order volume, customer counts, and product counts
Customer Analysis	Ordering frequency, repeat behavior, and customer loyalty patterns
Product Analysis	Most purchased, most reordered, and rarely purchased products
Department & Aisle Analysis	Sales contribution by department and aisle
Time-Based Analysis	Ordering patterns by day of week and hour of day
Advanced Business Analysis	Multi-step behavioral and ranking analysis using window functions

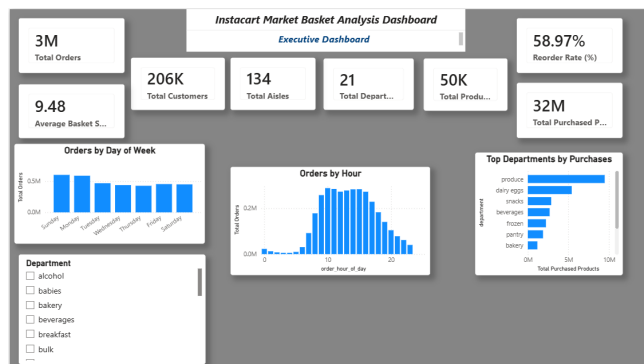
Key SQL Techniques Used

The queries make use of JOINS and SELF JOINS, GROUP BY with aggregate functions, Common Table Expressions (CTEs), and window functions including ROW_NUMBER(), RANK(), LAG(), and SUM() OVER() — allowing ranked product lists, customer-level running metrics, and period-over-period comparisons to be produced directly in SQL ahead of the Power BI layer.

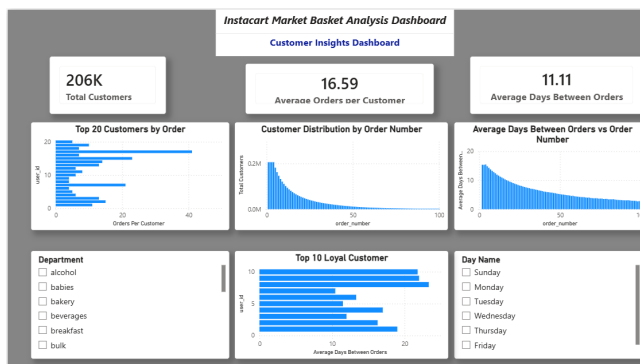
Together, these queries validate and extend the findings surfaced in the Power BI dashboard, providing a query-level foundation for the department, product, and time-based metrics presented on the following page.

Power BI Dashboard & Key Insights

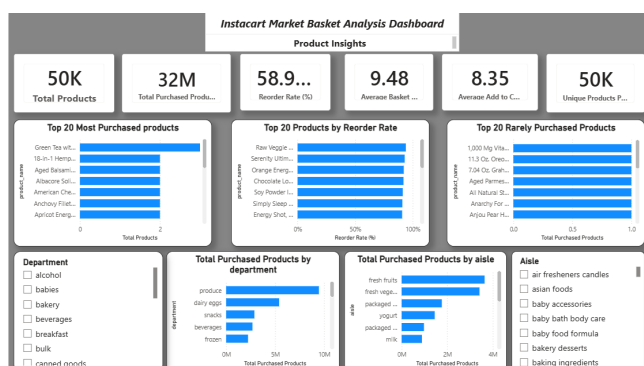
The insights above were modeled in Power BI into a 4-page interactive dashboard, built with DAX measures over the cleaned data model.



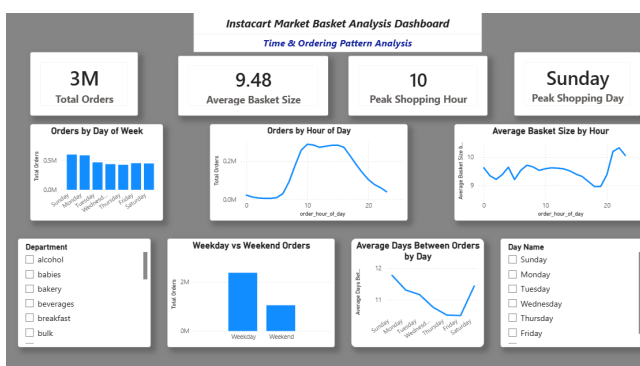
1. **Executive Dashboard** — Total orders, customers, products, average basket size, and reorder rate at a glance.



2. **Customer Insights** — Customer ordering frequency, top and most loyal customers, and days between orders.



3. **Product Insights** — Top purchased and reordered products, rarely purchased items, and department/aisle sales.



4. **Time & Ordering Pattern Analysis** — Orders by day and hour, average basket size by hour, and weekday vs. weekend behavior.

Key Findings

- Across 3M orders from 206K customers, the average basket size is 9.48 items, with an overall reorder rate of 58.97%, indicating strong repeat-purchase behavior.
- Customers place an average of 16.59 orders each, with an average of 11.11 days between orders; order frequency and the gap between orders both decline as customers place more orders.
- The Produce and Dairy & Eggs departments generate the highest purchase volume, followed by Snacks, Beverages, and Frozen — Fresh Fruits and Fresh Vegetables are the top aisles.
- Sunday is the peak shopping day and 10 AM is the peak shopping hour, with order volume concentrated between roughly 8 AM and 6 PM; weekday order volume exceeds weekend volume.
- Average basket size dips in the early morning hours and rises again late at night, while staying relatively stable through the main daytime shopping window.

Business Recommendations & Conclusion

Recommendations

Area	Recommendation
Customer Retention	Target customers whose order frequency is declining with re-engagement offers, since order gaps wider than 30 days indicate a high risk of churn.
Personalized Recommendations	Use individual reorder history to power personalized "buy again" suggestions, leveraging the high overall reorder rate.
Inventory Optimization	Prioritize stock availability for high-volume departments such as Produce and Dairy & Eggs, and allocate more space to slow-moving items in the aisles.
Targeted Marketing	Schedule promotions and notifications around the peak shopping day (Sunday) and peak hour (10 AM) to maximize reach.
Product Bundling	Bundle frequently reordered products with rarely purchased items in the same aisle to lift visibility of slow-moving products.
Peak-Time Operations	Align staffing, fulfillment, and delivery slot capacity with the daytime concentration of order volume seen on weekends.
Reorder-Based Marketing	Trigger reminders timed to a customer's typical day gap between orders to reduce churn between purchases.

Conclusion

RetailIQ demonstrates a complete, end-to-end analytics workflow on the Instacart Market Basket Analysis dataset — starting from business and data understanding, through Python-based cleaning and EDA, 25+ SQL business queries, and a governed Power BI data model, into a 4-page interactive executive dashboard. The resulting insights on customer behavior, product performance, and time-based ordering patterns translate directly into practical retention, marketing, and operational recommendations, illustrating how raw transactional data can be turned into decision-ready business intelligence.

Prepared by Anirban Sur | Tools: Python, SQL, Power BI